

East West University
Department of Mathematical and Physical Sciences
Course Title: Coordinate Geometry and Vector Analysis, Course code: MAT 104
Semester: Fall 2025, Section: 13
Quiz Test-I, Total Marks: 10 (3+3+4), Duration: 30 min.
Answer all the questions

1. By transforming the parallel axes through a properly chosen point (h, k) , prove that the equation $2x^2 - 11xy + 2y^2 + 6x - 5y + 2 = 0$ can be reduced to one containing only terms of the second degree. Find (h, k) .
2. Show that the straight lines represented by $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ are perpendicular to each other and $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ are parallel.
3. Define direction ratios. $n=3$
Establish the condition under which the straight line whose direction cosines are given by the relation $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2$ are perpendicular.

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Course

and vector Analysis

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Quiz Test-III, Total Marks: 10, Duration: 30 min.

Answer all the questions

1. Prove that $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{C}(\vec{B} \cdot \vec{D} \times \vec{A}) - \vec{D}(\vec{A} \cdot \vec{B} \times \vec{C})$. 3
Write the area of the triangle. 1
2. If $\vec{F} = (5xy - 6x^2)\hat{i} + (2y - 4x)\hat{j} - 4x^2 z^2 \hat{k}$, find the scalar potential of $\vec{F}$. 2
Define the circulation of the velocity of a liquid round the closed curve. 1
3. If $\vec{A} = x^2 y \hat{i} - 2xz \hat{j} + 2yz \hat{k}$. Find $\text{Curl Curl } \vec{A}$. 3



Mid-Term Examination
Course Title: Coordinate Geometry and Vector Analysis
Course Code: MAT 104 (Section: 13), Instructor: MYESMIN
Semester: Fall' 25

Time: 1 hour 10 min.

Full Marks: 25

-: Answer any of the following five (5) questions:-

1. (a) Define Invariants. [1]
(b) Transform the equation $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$ to one in which there is no term involving xy . [4]
2. (a) State the condition under which the pair of straight lines is real and coincident. [1]
(b) Show that the lines joining the origin to the intersection of $x^2 + y^2 + 19x + 4y - 3 = 0$ and $3x + 4y = 1$ are at right angles [4]
3. A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that,
 $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$. [5]
4. (a) Write down the symmetrical form of the straight line. [1]
(b) Find the direction cosines of the shortest distance between the lines
 $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$. [4]
Derive the direction ratios of that line.
5. The circle $x^2 + y^2 - 2x + 4y - 20 = 0$ meets the straight line $2x - y = 2$ at A and B . Find the equation of the straight line joining A and B with the origin and obtain the angle between them. [5]

Good Luck